

Review retrospective:  
*the J-function sextic, a refactored twin that left its sibling  
behind*

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### Abstract

A soundness review of `Laplace/OneD/JFunctionSextic.lean`: the  $J$ -function asymptotic for the sextic,  $t^{1/6} \int e^{-tw^6/720} \pi(w) dw \rightarrow \pi(0) \cdot \frac{1}{3} \cdot 720^{1/6} \Gamma(1/6)$ , now a thin  $k = 3$  specialisation of the generic `kth_jfunction_*` family. Result: **a clean fidelity pass with three small cleanups** — a dead private bridge lemma, a dead `open Real`, and a `show→change` style fix — plus a report-only observation that the quartic twin was never migrated to the same refactor.

## 1 Setting

Unlike the quartic  $J$ -function (still a standalone substitute-then-DCT proof), the sextic file was refactored (its “L3” tide) to be a five-line specialisation of the generic- $k$   $J$ -function abstraction, inheriting the mathematically-minimal hypothesis class (`AE measurable + ContinuousAt 0 + global bound`). It is the cleanest member of the family to review — the content lives upstream, and the file is just the  $k = 3$  constant bridging.

## 2 What was reviewed

Two public theorems (the centred DCT core and the asymptotic corollary), each a  $k = 3$  instance of the generic theorems, plus three private constant-bridge lemmas ( $2 \cdot 3 = 6$ ,  $6! = 720$ ,  $1/((2 \cdot 3:\mathbb{N}):\mathbb{R}) = 1/6$ ).

## 3 Fidelity / soundness findings

**Sound; no mismatch.** The bridges are correct ( $6! = 720$ ,  $1/6$ ), the scaling exponent is  $t^{1/6}$ , the kernel is  $e^{-tw^6/720}$ , and the limit  $\pi(0) \cdot \frac{1}{3} \cdot 720^{1/6} \Gamma(1/6)$  is exactly the parametric  $\pi(0) \cdot (1/k) \cdot ((2k)!)^{1/(2k)} \Gamma(1/(2k))$  at  $k = 3$  (the prefactor is  $1/k = 1/3$ , not  $1/(2k)$ ). The weakened hypotheses are carried through faithfully — no global `Continuous` is reintroduced.

One report-only observation: the quartic  $J$ -function file still uses its original standalone proofs with the *stronger* `Continuous` prior hypothesis and was *not* migrated to specialise the generic family the way the sextic was — even though the family’s L1 retrospective explicitly anticipated “both existing files become 5-line specialisations”. So the two siblings are now inconsistent: sextic on the minimal hypotheses, quartic on the global one. That is a forward-tide follow-up (migrate the quartic), not a review edit, and it corroborates the hypothesis note raised in the quartic review.

## 4 Simplifications applied

**Three.** A dead private lemma `sextic_kth_eq_const_six` ( $(2 \cdot 3:\mathbb{N}) = 6$ ) — unused, because the  $2 \cdot 3 \rightarrow 6$  passage is handled definitionally rather than through the named lemma; a dead `open Real` (every `Real.exp/Real.Gamma` is qualified, and `open MeasureTheory/Filter/Topology` carry the load — the last via  $\mathcal{N}$ ); and a `show→change` on the factorial-bridge lemma, clearing a standing `linter.style.show` warning per the seabed’s “change, not show” rule. Rebuild-arbitrated green, file now warning-free, both public signatures byte-identical, GPT-affirmed. (A residual `show`-style warning lives in the upstream `JFunctionMonomial.lean` — out of scope here, flagged for that file’s review.)

## 5 Disagreements and open questions

None on the mathematics.

## 6 Lessons

- **A refactor can migrate one sibling and silently strand the other.** The J-function family’s hypothesis-weakening refactor reached the sextic but not the quartic, leaving two files that prove the “same” kind of theorem under different hypotheses. A review of either alone would miss it; reviewing them in succession surfaced the asymmetry. When a file advertises itself as “the  $k = c$  specialisation” of a generic, check whether its siblings made the same move.
- **Three small file-local cleanups, three independent arbiters.** The dead private lemma (grep: unused), the dead open (grep: no notation used), and the `show/change` (linter: style warning) are each confirmed by a different mechanical check, and the single rebuild settles all three at once with no new warnings.

## 7 Follow-ups

- **Forward tide:** migrate `JFunctionQuartic` to specialise the generic `kth_jfunction_*` (as the sextic does), weakening its hypothesis from `Continuous` to `AMeasurable + ContinuousAt 0`.
- **Next review:** the generic `JFunctionMonomial.lean` (the  $N1$  abstraction both J-function files rest on) — it also carries a `show`-style warning to address. Then `JFunctionAsymptoticEquivalence`.